Exercise 1 — Exploratory Data Analysis of MILK10k

Computer Vision & Speech Recognition · Joshua Cole · 08 November 2026

1. Public GitHub repository

https://github.com/jcole-jpg/Computer-Vision-and-Speech-Recognition

EDA code: `notebooks/01_eda_milk10k.ipynb` (executed) · helpers: `src/milk10k/` · figures: `reports/figures/`

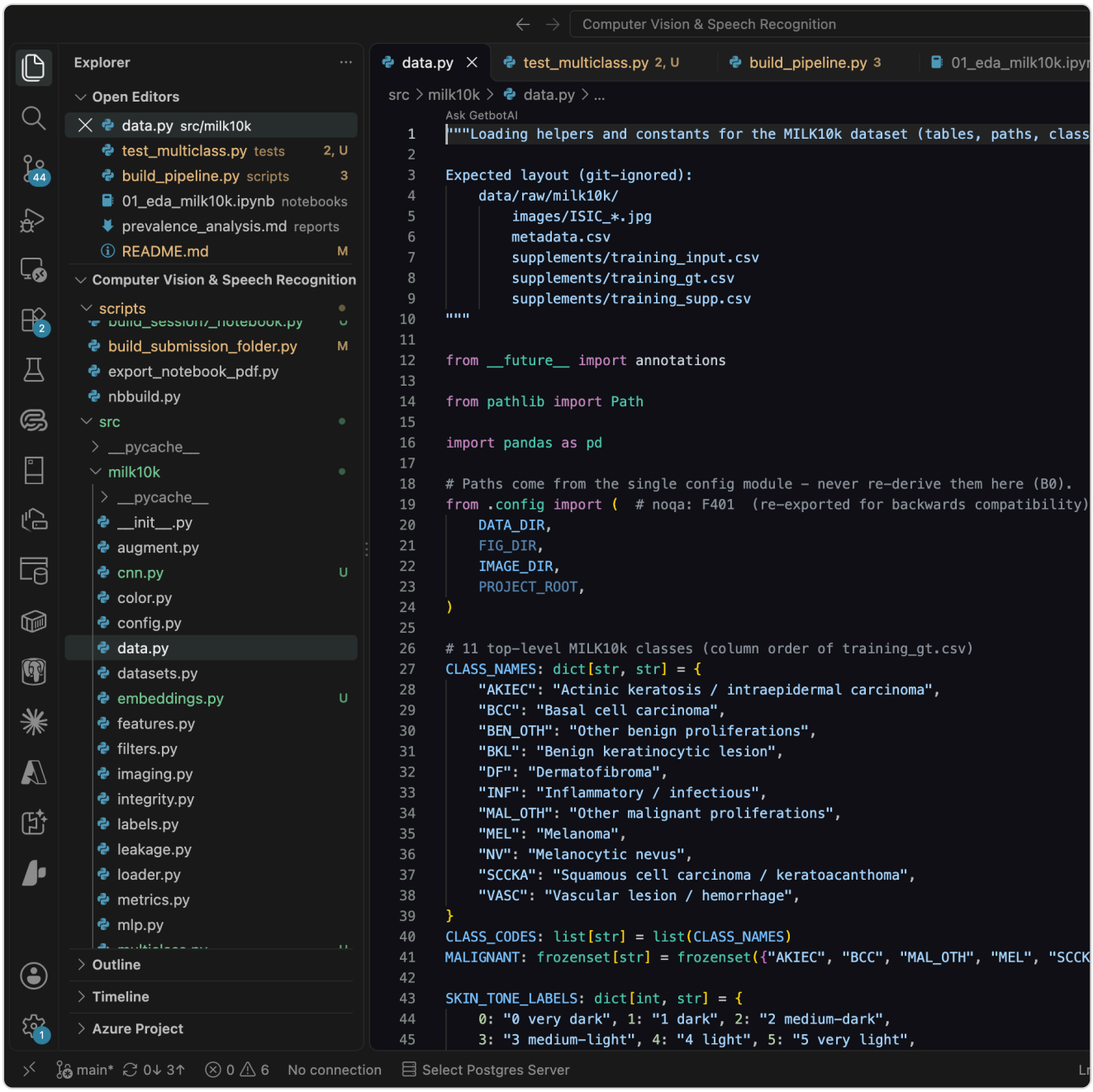
2. Tasks covered

Task	Where
Complete the EDA of MILK10k: images, metadata, class distribution, visualisations	`notebooks/01_eda_milk10k.ipynb` · 12 figures in `reports/figures/`
Choose an IDE / code editor	VS Code (screenshot, section 4)
Create a GitHub repository for course work and project	Public repo linked above
Create a project structure and push the EDA exercises	Layout on page 3; all code pushed to `main`
Small explanation of the clinical task	Section 3 below and `README.md` §1

3. Clinical task

Classify a skin lesion into its diagnostic category from a **paired clinical close-up and dermatoscopic image** plus patient metadata (age, sex, site, skin tone), so that malignant lesions (melanoma, basal cell carcinoma, squamous cell carcinoma, …) are flagged for biopsy and benign ones are not over-treated. MILK10k provides 5,240 lesions / 10,480 images with 11 top-level classes (48 fine diagnoses); ~96 % of labels are histopathology-confirmed.

4. Project structure (code editor: VS Code)



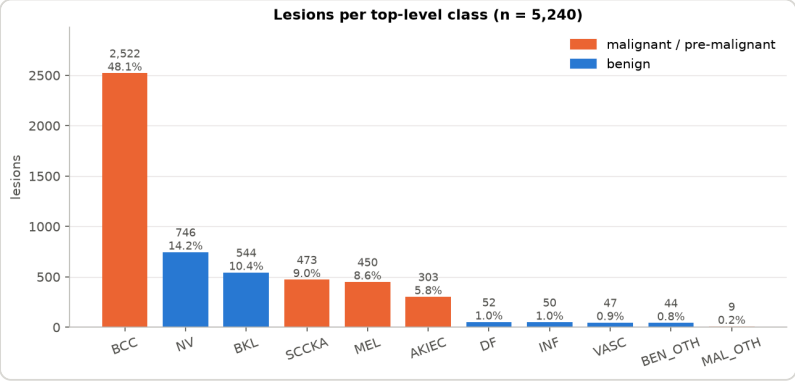
Screenshot of the project Explorer in VS Code.

5. Directory layout

```
Computer-Vision-and-Speech-Recognition/
├── README.md                    project overview, clinical task, dataset, findings
├── SUBMISSION.md                link list: every deliverable and where it lives
├── requirements.txt             Python dependencies
├── .gitignore                   keeps data / venv / checkpoints out of git
├── data/
│   ├── raw/milk10k/             extracted dataset (git-ignored)
│   └── processed/               derived tables (lesions.csv, hw_*.csv)
├── notebooks/                   01_eda_milk10k, 02_image_basics, 03_eda_pipeline,
│                                04_filters, 05_classical_baseline, 06_*  (executed)
├── src/milk10k/                 data, labels, splits, imaging, preprocess, loader,
│                                datasets, viz, color, features, filters, mlp ...
├── scripts/                     build_pipeline.py, notebook builders, PDF export
├── tests/                       pytest suite for the package
├── reports/                     Markdown reports + figures/
├── exercises/                   exercise PDFs (exercise_1 = this deliverable)
└── docs/
    ├── build_deliverable.py     builds exercises/exercise_1.pdf
    └── editor_screenshot.png
```

6. EDA summary

Finding	Implication for modelling
Exactly 1 clinical + 1 dermatoscopic image per lesion	Split by `lesion_id`; model the pair jointly
72 % of lesions malignant; BCC alone 48 %, nevi only 14 % — biopsy-enriched cohort	Use macro-F1 / balanced accuracy / per-class sensitivity, not accuracy
Five classes with < 55 lesions (MAL_OTH has 9)	Stratified splits, class weighting / re-sampling
48 fine diagnoses, most with < 10 lesions	Train at the 11-class level
Median age 65, 60 % male, 61 % skin-tone 3, < 1 % tones 0–1	Report metrics per tone / age band
Age, site and skin tone vary systematically by class	Legitimate priors but possible shortcuts — audit
Mixed resolutions; dermatoscopy brighter, with gel / hair / ink artefacts	Per-modality normalisation and augmentation



Lesions per top-level class (n = 5,240), coloured by malignancy. Full analysis in the notebook.